MIDTERM REVIEW
CONNECT THE IDEAS

BOONE BOWLES | NOVEMBER 5, 2026

1 | Firms and value | Remember

- Legal form affects liability, taxation, continuity and access to capital.
- Ownership and control can be separate. Incentives and oversight matter.
- Bigger earnings, more sales or a larger firm do not automatically mean more value.
- Ask what creates durable cash flows, and how stakeholders affect them.

1 | Firms and value | Try it

1. A CEO rejects a valuable long-term investment to protect this year's bonus. What is the main concern?

A. Too little depreciation B. An agency conflict C. A shortage of shareholders

2. Which feature most directly helps a corporation pool capital at scale?

A. Transferable ownership shares B. A guarantee of positive profits C. Every owner manages operations

3. An LLC is taxed as a partnership. Which statement fits our class comparison?

A. Its owners necessarily have unlimited liability B. Its profits must face C-corporation double taxation C. Liability protection and pass-through taxation can coexist

1 | Firms and value | Answers

1 | B The pay rule rewards a personal objective that can conflict with long-term owner value.

2 | A Many investors can hold and transfer claims without running day-to-day operations.

3 | C Legal liability and tax treatment are different dimensions. The question explicitly specifies partnership taxation.

2 | Statements and ratios | Remember

- Balance sheet: a date. Income statement and cash-flow statement: a period.
- Compare like dates, units and definitions; book value and market value differ.
- DuPont separates profitability, asset efficiency and the equity denominator.
- Profit is not cash flow: investment in fixed assets and working capital uses cash.

2 | Statements and ratios | Try it

1. Which statement reports the amount of inventory held on a particular date?

A. Income statement B. Balance sheet C. Statement of cash flows

2. Two firms have equal ROE. What does DuPont tell us?

A. Their profit margins must match B. Their asset turnover must match C. Different combinations of the three drivers can produce the same ROE

3. A firm reports positive net income but spends heavily on equipment and inventory. Its FCF can be:

A. Negative B. Necessarily equal to net income C. Positive by definition

2 | Statements and ratios | Answers

1 | B Inventory is an asset balance at a date. Its purchase and sale affect other statements over time.

2 | C Equal products do not require equal factors. Identify which driver differs before interpreting performance.

3 | A Cash investment can exceed after-tax operating cash generation.

Walmart / Target | Calculate first

1. Using the latest-year figures, which firm has the higher current ratio?

A. Walmart, about 1.27× B. Target, about 0.94× C. Target, about 1.06×

2. Which description fits the latest-year net profit margin and asset turnover?

A. Walmart has the higher margin and lower turnover B. Target has the lower margin and higher turnover C. Walmart has the lower margin and higher turnover

3. Target's operating-income coverage of net interest changed from the prior year to the latest year by:

A. Falling from about 13.54× to 11.50× B. Rising from about 11.50× to 13.54× C. Remaining unchanged

Walmart / Target | Answers 1–3

1 | B Walmart: 84,874 / 107,469 = 0.790×. Target: 20,005 / 21,230 = 0.942×. Neither number alone establishes financial distress.

2 | C Margins: Walmart 21,893 / 713,163 = 3.07%; Target 3,705 / 104,780 = 3.54%. Turnover: 2.51× versus 1.76×, using year-end assets.

3 | A 5,566 / 411 = 13.54×; 5,117 / 445 = 11.50×. Operating income fell while net interest expense rose.

Walmart / Target | Interpret the ratios

4. Which firm had the higher latest-year gross margin using the worksheet's revenue convention?

A. Walmart, about 24.93% B. Target, about 27.93% C. Walmart, about 75.07%

5. Walmart's revenue increased from 680,985 to 713,163. Approximate growth was:

A. 4.73% B. 95.49% C. 3.07%

6. Can these ratios alone establish which stock is the better investment?

A. Yes: choose the higher profit margin B. Yes: choose the higher asset turnover C. No: price, expected future cash flows and risk also matter

Walmart / Target | Answers 4–6

4 | B Walmart: (713,163 – 535,395) / 713,163 = 24.93%. Target: (104,780 – 75,511) / 104,780 = 27.93%. Cost classifications differ across retailers.

5 | A (713,163 – 680,985) / 680,985 = 4.73%. Use the prior-year amount in the denominator.

6 | C Historical ratios describe operations and financial position. Investment value also depends on what you pay and what happens next.

3 | Time, bonds and interest rates | Remember

- Draw dates first. Move every cash flow to the same valuation date.
- Match periods: semiannual coupon, semiannual yield, twice as many periods.
- For fixed promised payments: a higher required yield means a lower price.
- Risk structure compares similar maturities; term structure compares maturities.

3 | Time, bonds and interest rates | Try it

1. A $1,000 bond pays a 5% annual coupon semiannually and has 7 years left. At 6.5% annual YTM, what is its price just after a coupon?

A. $953.71 B. $916.71 C. $917.73

2. You receive $5,000 at each year-end in Years 4, 5 and 6, and nothing else. At 8% annually, what is the value today?

A. $12,885.48 B. $9,471.22 C. $10,228.91

3. Investors view a particular bond as less risky, with other influences fixed. What happens?

A. Demand rises; price rises; yield falls B. Supply rises; price falls; yield rises C. The coupon rate rises automatically

3 | Time, bonds and interest rates | Answers

1 | B N = 14; rate per half-year = 3.25%; PMT = $25; FV = $1,000. Solve PV = –$916.71: the quoted price is $916.71. Use END mode and P/Y = C/Y = 1 for these period inputs.

2 | C First find the Year 3 value: N = 3, I/Y = 8, PMT = 5,000, FV = 0 → $12,885.48. Discount it three more years: 12,885.484936 / 1.08^3 = $10,228.91. Use END mode.

3 | A The demand curve shifts right. The promised coupon stays fixed; price adjusts and yield moves inversely.

Interest rates | Two more checks

1. An upward yield curve with a positive maturity premium proves future short rates will rise. True or false?

A. True B. False C. Only if the bond pays coupons

2. Expected inflation rises, holding other drivers fixed. In the market for nominal bonds:

A. Demand shifts right; supply left B. Both curves shift left C. Demand shifts left; supply right

Interest rates | Answers

1 | B False. A positive term premium can create an upward slope even with flat expected short rates.

2 | C Investors expect lower real returns and borrowers face lower expected real repayment costs. Price falls and nominal yield rises.

One bond | Three prices

- $1,000 face value; 6% annual coupon; semiannual payments; six years remaining.
- 1. Today: annual YTM = 8%. Price the bond.
- 2. Two years later, just after coupon 4: YTM is still 8%. Price again.
- 3. At that same date: YTM falls to 5%. Price again.

Bond price 1 | Today

- N = 12 half-years; PMT = $30; FV = $1,000; yield per period = 4%.

- Price = $906.15. Coupon rate 6% < market yield 8%: a discount bond.

- Calculator: N 12, I/Y 4, PMT 30, FV 1000 → PV −906.15.

Bond price 2 | Two years later

- Four coupons have been paid. Eight remain: N = 8, PMT = $30, FV = $1,000.
- YTM stays 8% annually → 4% per half-year. Price = $932.67.
- At unchanged yield, the discount shrinks as maturity approaches.

Bond price 3 | Same date, lower yield

- N is still 8; PMT is still $30; FV is still $1,000.
- New annual YTM = 5% → 2.5% per half-year. Price = $1,035.85.
- The lower discount rate raises price. Coupon 6% > yield 5%: now a premium.

4 | Project decisions and cash flows | Remember

- Compare with versus without the project. For replacement, use replace minus keep.
- Exclude sunk costs; include opportunity costs and lost contribution.
- FCF = NOPAT – ΔNFA – ΔNWC.

- Use NPV to choose feasible alternatives; test timing, inputs and terminal cash flows.

4 | Project decisions and cash flows | Try it

1. Depreciation rises by $100, with cash revenue and cash costs unchanged. At a 25% tax rate, operating FCF:

A. Falls $100 B. Is unchanged C. Rises $25

2. A machine sells for $30,000; tax book value is $40,000. At 25% tax, with usable losses, after-tax proceeds are:

A. $32,500 B. $27,500 C. $40,000

3. Required NWC rises from $12,000 to $17,000. What enters this period's FCF?

A. −$17,000 B. −$5,000 C. +$5,000

4 | Project decisions and cash flows | Answers

1 | C The noncash expense is added back, while the $25 tax saving remains. Assume the deduction can be used.

2 | A $30,000 − 25% × ($30,000 − $40,000) = $32,500. The loss creates a $2,500 tax benefit.

3 | B Subtract the $5,000 increase. The ending balance itself is not this period's cash investment.

8.15 | Beryl’s choice

- Keep renting: $50,000 a year, maintenance included.
- Buy current: $150,000 now; $20,000 annual maintenance.
- Buy advanced: $250,000 now; $15,000 annual maintenance; $10,000 annual bottling savings; $35,000 training now.
- Both: 10-year life; 7-year straight-line depreciation; no salvage; tax 35%; required return 8%.

8.15 | Set it up, then choose

- Use continuing to rent as the baseline. Its incremental NPV is zero.
- Work in pairs: initial outlay, Years 1–7 FCF, Years 8–10 FCF.
- Why does cash flow change after Year 7? Where does training belong?
- Compare NPVs. Does the more advanced machine necessarily create more value?

8.15 | Buy the current machine

- Current: −$150,000 now; $27,000 in Years 1–7; $19,500 in Years 8–10.
- Advanced: −$272,750 now; $41,750 in Years 1–7; $29,250 in Years 8–10.
- NPVs: rent $0; current +$19,894; advanced −$11,400.
- Choose the current machine. Larger operating savings do not offset every larger investment.

8.25 | Billingham's expansion

- Machine: $2.75m now; 10-year straight-line depreciation; no salvage. The $50,000 study is already paid.
- Normal new sales: $10m/year. Our class version loses $5m of those sales in Year 1.
- COGS: 70% of net incremental sales. Added selling/admin: $2m/year.
- Tax 15%; required return 10%; tax losses immediately usable.

8.25 | Working capital and the timeline

- Inventory: $1m at Year 0, held through Year 9.
- At year-ends 1–9: receivables = 15% of sales; payables = 10% of COGS.
- Recover all NWC at the end of Year 10. Subtract changes, not balances.
- Start with Years 0, 1, 2 and 10. Then fill the repeated operating years.

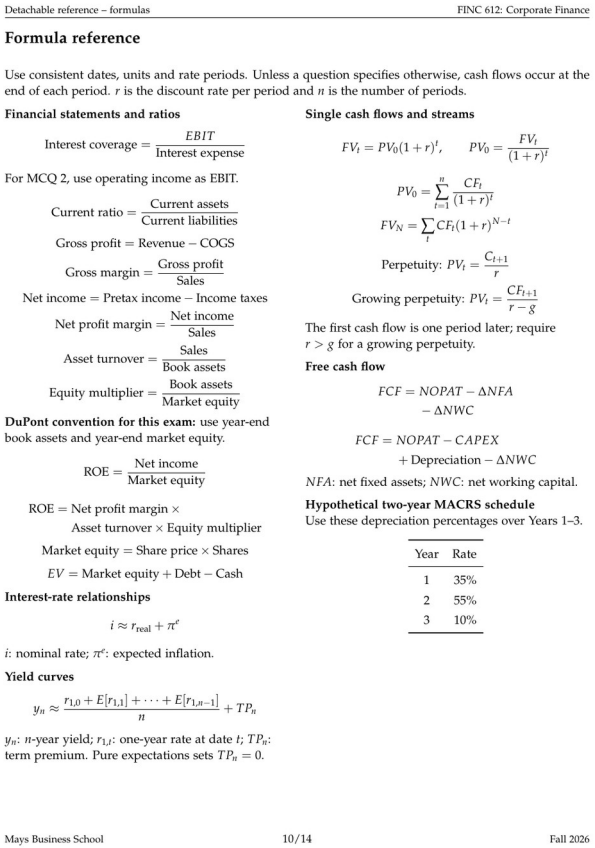
8.25 | Check the cash flows

- Year 0: −$3,750,000 = machine + inventory investment.
- Year 1: −$783,750. Year 2: +$491,250.
- Years 3–9: +$891,250 each. Year 10: +$2,691,250.
- NPV = +$567,018 → accept under these assumptions.

What the exam pages ask you to do

- Read the narrative, build the cash flows and use the checkpoints to organize your work.
- Distinguish your project calculation from any separate Cedar example printed in a question.
- Select the requested T/F or MCQ response. Blanks and model grids are working space.
- Use the supplied formula sheet’s definitions, dates and depreciation schedule.

The reference sheet | Know where to look



Detachable reference – formulas FINC 612: Corporate Finance

Formula reference

Use consistent dates, units and rate periods. Unless a question specifies otherwise, cash flows occur at the end of each period. r is the discount rate per period and n is the number of periods.

Financial statements and ratios

$$\text{Interest coverage} = \frac{EBIT}{\text{Interest expense}}$$

For MCQ 2, use operating income as EBIT.

$$\text{Current ratio} = \frac{\text{Current assets}}{\text{Current liabilities}}$$

$$\text{Gross profit} = \text{Revenue} - \text{COGS}$$

$$\text{Gross margin} = \frac{\text{Gross profit}}{\text{Sales}}$$

$$\text{Net income} = \text{Pretax income} - \text{Income taxes}$$

$$\text{Net profit margin} = \frac{\text{Net income}}{\text{Sales}}$$

$$\text{Asset turnover} = \frac{\text{Sales}}{\text{Book assets}}$$

$$\text{Equity multiplier} = \frac{\text{Book assets}}{\text{Market equity}}$$

DuPont convention for this exam: use year-end book assets and year-end market equity.

$$\text{ROE} = \frac{\text{Net income}}{\text{Market equity}}$$

$$\text{ROE} = \text{Net profit margin} \times \text{Asset turnover} \times \text{Equity multiplier}$$

$$\text{Market equity} = \text{Share price} \times \text{Shares}$$

$$EV = \text{Market equity} + \text{Debt} - \text{Cash}$$

Interest-rate relationships

$$i \approx r_{\text{real}} + \pi^e$$

i: nominal rate; π^e: expected inflation.

Yield curves

$$y_n \approx \frac{r_{1,0} + E[r_{1,1}] + \cdots + E[r_{1,n-1}]}{n} + TP_n$$

y_n: n-year yield; $r_{1,t}$: one-year rate at date t; TP_n: term premium. Pure expectations sets $TP_n = 0$.

Single cash flows and streams

$$FV_t = PV_0(1+r)^t, \qquad PV_0 = \frac{FV_t}{(1+r)^t}$$

$$PV_0 = \sum_{t=1}^{n} \frac{CF_t}{(1+r)^t}$$

$$FV_N = \sum_t CF_t(1+r)^{N-t}$$

$$\text{Perpetuity: } PV_t = \frac{C_{t+1}}{r}$$

$$\text{Growing perpetuity: } PV_t = \frac{CF_{t+1}}{r-g}$$

The first cash flow is one period later; require $r > g$ for a growing perpetuity.

Free cash flow

$$FCF = NOPAT - \Delta NFA - \Delta NWC$$

$$FCF = NOPAT - CAPEX + \text{Depreciation} - \Delta NWC$$

NFA: net fixed assets; NWC: net working capital.

Hypothetical two-year MACRS schedule

Use these depreciation percentages over Years 1–3.

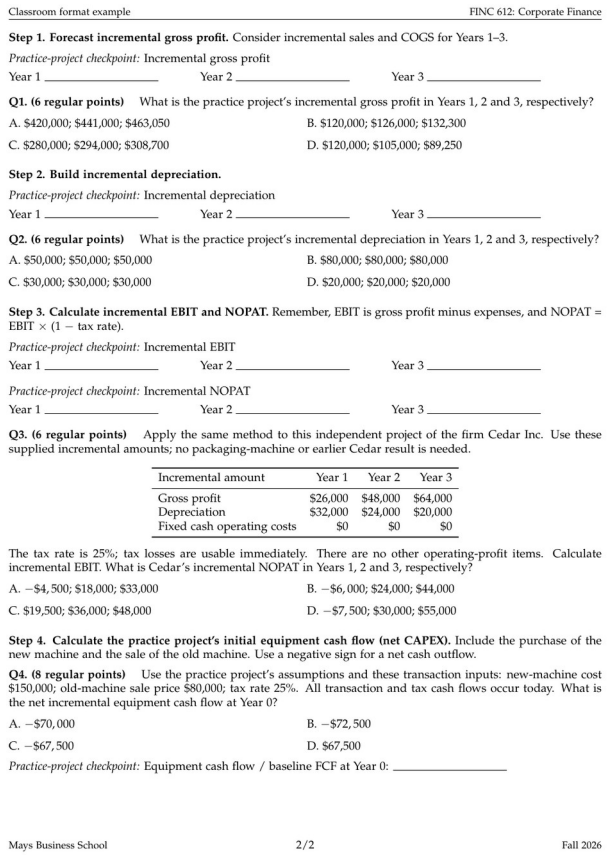
Year	Rate
1	35%
2	55%
3	10%

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Formula sheet + blank modeling grids + working paper.

Read each formula's definitions. The exam specifies market equity for DuPont / ROE.

The classroom example | Setup and checkpoints



Classroom format example FINC 612: Corporate Finance

Step 1. Forecast incremental gross profit. Consider incremental sales and COGS for Years 1–3.

Practice-project checkpoint: Incremental gross profit

Year 1 _____ Year 2 _____ Year 3 _____

Q1. (6 regular points) What is the practice project's incremental gross profit in Years 1, 2 and 3, respectively?

A. $420,000; $441,000; $463,050 B. $120,000; $126,000; $132,300

C. $280,000; $294,000; $308,700 D. $120,000; $105,000; $89,250

Step 2. Build incremental depreciation.

Practice-project checkpoint: Incremental depreciation

Year 1 _____ Year 2 _____ Year 3 _____

Q2. (6 regular points) What is the practice project's incremental depreciation in Years 1, 2 and 3, respectively?

A. $50,000; $50,000; $50,000 B. $80,000; $80,000; $80,000

C. $30,000; $30,000; $30,000 D. $20,000; $20,000; $20,000

Step 3. Calculate incremental EBIT and NOPAT. Remember, EBIT is gross profit minus expenses, and NOPAT = EBIT × (1 − tax rate).

Practice-project checkpoint: Incremental EBIT

Year 1 _____ Year 2 _____ Year 3 _____

Practice-project checkpoint: Incremental NOPAT

Year 1 _____ Year 2 _____ Year 3 _____

Q3. (6 regular points) Apply the same method to this independent project of the firm Cedar Inc. Use these supplied incremental amounts; no packaging-machine or earlier Cedar result is needed.

Incremental amount	Year 1	Year 2	Year 3
Gross profit	$26,000	$48,000	$64,000
Depreciation	$32,000	$24,000	$20,000
Fixed cash operating costs	$0	$0	$0

The tax rate is 25%; tax losses are usable immediately. There are no other operating-profit items. Calculate incremental EBIT. What is Cedar's incremental NOPAT in Years 1, 2 and 3, respectively?

A. −$4,500; $18,000; $33,000 B. −$6,000; $24,000; $44,000

C. $19,500; $36,000; $48,000 D. −$7,500; $30,000; $55,000

Step 4. Calculate the practice project's initial equipment cash flow (net CAPEX). Include the purchase of the new machine and the sale of the old machine. Use a negative sign for a net cash outflow.

Q4. (8 regular points) Use the practice project's assumptions and these transaction inputs: new-machine cost $150,000; old-machine sale price $80,000; tax rate 25%. All transaction and tax cash flows occur today. What is the net incremental equipment cash flow at Year 0?

A. −$70,000 B. −$72,500

C. −$67,500 D. $67,500

Practice-project checkpoint: Equipment cash flow / baseline FCF at Year 0: _____

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Read the stem: your project result, or a separate Cedar example?

Use the checkpoints to organize your work; select the requested answer.

Before tomorrow’s exam

- Practice setting up the timeline and calculator without Excel.
- Check: units, dates, signs, rate periods, tax effects and ending cash flows.
- Use tonight’s TA feedback to target your remaining gaps.
- What is the one step you want to clarify before we finish?